Combined Gestational Age- and Birth Weight–Adjusted Cutoffs for Newborn Screening of Congenital Adrenal Hyperplasia

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Context: Congenital adrenal hyperplasia (CAH) was among the first genetic disorders included in newborn screening (NBS) programs worldwide, based on 17α-hydroxyprogesterone (17-OHP) levels in dried blood spots. However, the success of NBS for CAH is hampered by high false positive (FP) rates, especially in preterm and low-birthweight infants.

Objective: To establish a set of cutoff values adjusting for both gestational age (GA) and birth-weight (BW), with the aim of reducing FP rates.

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Abbreviations: 17-OHP, 17α-hydroxyprogesterone; BW, birthweight; CAH, congenital adrenal hyperplasia; DBS, dried blood spots; FP, false-positive; GA, gestational age; NBS, newborn screening; NICU, neonatal/preterm intensive care unit; PPV, positive predictive value.

Design: This cross-sectional, population-based study summarizes 10 years of experience of the Israeli NBS program for diagnosis of CAH. Multitiered 17-OHP cutoff values were stratified according to both BW and GA.

Participants: A total of 1,378,132 newborns born between 2008 and 2017 were included in the NBS program.

Results: Eighty-eight newborns were ultimately diagnosed with CAH; in 84 of these, CAH was detected upon NBS. The combined parameters–adjusted approach significantly reduced the recall FP rate (0.03%) and increased the positive predictive value (PPV) (16.5%). Sensitivity among those referred for immediate attention increased significantly (94%). There were four false negative cases (sensitivity, 95.4%), all ultimately diagnosed as simple-virilizing. Sensitivity and specificity were 95.4% and 99.9%, respectively, and the percentage of true-positive cases from all newborns referred for evaluation following a positive NBS result was 96%.

Conclusions: The use of cutoff values adjusted for both GA and BW significantly reduced FP rates (0.03%) and increased overall PPV (16.5%). Based on our 10 years of experience, we recommend the implementation of this two parameter–adjusted approach for NBS of classic CAH in NBS programs worldwide. ***(J Clin Endocrinol Metab* 104: 3172–3180, 2019)**

Congenital adrenal hyperplasia (CAH) is a group of autosomal recessive disorders caused by deficiencies of steroidogenic enzymes leading to inborn errors of cortisol biosynthesis. More than 90% of cases are caused by deficiency of 21-hydroxylase, associated with deleterious variants in the *CYP21A2* gene (1). CAH is divided into two groups and three main clinical presentations depending on the type of mutation and the level of enzymatic activity (2, 3). Seventy-five percent of patients with the classic, more severe form have the salt-wasting type, and the remaining are designated as having the simple virilizing type (4). The worldwide incidence of the classic form of CAH is 1 per 10,000 to 1:15,000 live births (3). The classic form leads to prenatal virilization in the female fetus; hence, the diagnosis is made at birth. The other group is nonclassic CAH, defined also as late-onset, and so is less likely to be detected by newborn screening (NBS). Classic salt-wasting CAH is a life-threatening condition, especially in infancy, and early detection and therapeutic intervention can significantly alter the prognosis and reduce the associated morbidity and mortality (5). CAH has a relatively high incidence, and high-throughput screening tests are both sensitive and cost-effective (5). Thus, CAH has long been considered an excellent candidate for NBS and indeed was among the first genetic disorders to be included in NBS programs worldwide; screening is commonly based on the measurement of 17α-hydroxyprogesterone (17-OHP) levels in dried blood spots (DBS) (6–8).

Nevertheless, one of the greatest concerns hampering the success of NBS for CAH has been the high rate, up to 1%, of false-positive (FP) results (1). This caveat is particularly important in preterm and low-birthweight infants (9, 10), in part because of transient elevations in 17-OHP levels affected by birth stress, adrenal immaturity, and early collection of specimens before 24 hours (11).

Subsequently, over the past few decades, NBS worldwide have pursued several different strategies aimed at lowering the FP rates and improving the positive predictive value (PPV), with varying degrees of success.

One such approach has been to adjust the threshold levels of 17-OHP according to birthweight (BW) (12–14). Alternatively, cutoff levels based on gestational age were attempted and described (15, 16). In a study comparing the two determinants, van der Kamp *et al.* (17) used regression analysis and concluded that the latter was associated with greater specificity. Indeed, many countries worldwide chose to implement BW- or gestational age (GA)–adjusted cutoffs.

An additional approach, first implemented in Bavaria and described by Olgemöller *et al.* (18), based cutoff levels on both BW and age at sampling, subsequently improving the PPV and decreasing FP rates to 0.79%. Most recently, in a cohort of 271,810 newborns in Sao Paolo, Brazil, investigators based the cutoff values on BW and sample collection time (48 to 72 hours vs ≥72 hours) and reported a FP rate of 0.2% and 0.5% using the 99.8th and 99.5th percentiles, respectively. They concluded that the former was the best cutoff value to distinguish between affected and unaffected newborns (19).

Because the commonly used strategies did not collectively ameliorate the relatively high FP rates of NBS for CAH, second-tier testing was introduced with biochemical (liquid chromatography–tandem mass spectrometry, measuring adrenal steroid levels in DBS) or molecular (4, 20–24) methods. However, two-tier testing of a single sample resulted in an increased false negative (FN) rate, without significantly reducing the FP rate (23).

With the purpose of further reducing the FP rates of newborn screening for classic CAH, the National Newborn Screening Program in Israel has been implementing a unique set of cutoff values, combining adjustments for both BW and GA at birth. We hereby report our 10-year findings using this strategy.

Materials and Methods

The Newborn Screening Program in Israel is a national effort, and all samples are transferred to, handled, and analyzed at a single laboratory.

Preanalytical

Blood from neonates born in Israel is collected by a heel prick blotted on a 903 filter paper manufactured by Eastern Business Forms (Greenville, SC). Recommended collecting time is 36 to 72 hours from birth. DBS samples arrive to the laboratory for analysis within 3 to 5 days from birth. For newborns still hospitalized at 7 days of life, a second sample is obtained. The blood-collecting filter paper contains a text box in which to record clinically relevant information.

Additional data routinely reported electronically to the NBS program for each DBS sample, including GA rounded up to full weeks, sex, BW (in grams), and hospitalization department [nursery, neonatal/preterm intensive care unit (NICU)]. Mother's Israeli ID number, reported electronically, is used to automatically flag a sample with a positive family history.

Analytical

Levels of 17-OHP (in nmol/L) in DBS are measured by time-resolved fluoroimmunoassay by using the B024-112 AutoDELFIA® Neonatal 17α-hydroxyprogesterone kit (PerkinElmer Life and Analytical Sciences, Turku, Finland). The cutoffs were determined according to a pilot run of the program. The cutoff level of 17OHP is adjusted according to both BW and GA (Tables 1 and 2; Fig. 1). For the second sample, gestational age is corrected according to the date of sample collection.

Postanalytical

Positive results are reported both to the hospital of birth and to the community Mother-Child Health Clinic, based on electronically reported maternal address. Recall samples are routinely analyzed in our screening program. For an immediate referral of a discharged neonate, the newborn screening program nurse coordinates between the family and a pediatric endocrinologist in the nearby or family preferred hospital. A few weeks later, a follow-up survey is sent to the endocrinologist. The survey includes information on the diagnosis, type of CAH, family history, treatment modality, and genetic testing results.

The workflow described herein was based on the Israeli Ministry of Health, Government Regulation articles 17/2009 and 2/2018.

Pilot study

As the basis of the NBS strategy reported herein, we used GA and BW data of a pilot cohort of newborns, born in Israel and included in the National NBS program (n = 109,937). The initial cutoff of normal values was thus based on 17-OHP levels of 10,000 consecutive DBS tests (including for persons born in January 2008 included in this study, among which was the first diagnosed case of CAH, a male newborn designated as having

Table 1. Retrospective Analysis of NBS for CAH of 1,488,069 Newborns Based on 17-OHP Values for Percentiles Stratified by BW Alone, GA Alone, and Both BW and GA

			Percentile				
BW (g)[a]	GA (wk)[a]	Newborns (n = 1,488,069), n (%)	99.00th	99.90th	99.95th	99.97th	99.99th
<1500 g		15,575 (1.0)		288	320	357	
1500–1999		24,515 (1.6)		126	151	178	
2000–2499		83,594 (5.6)		85	104	123	
2500–2999		327,392 (22.0)		51	60	69	
3000–3499		611,117 (41.1)		32	39	45	
3500–3999		348,539 (23.4)		26	33	41	
≥4000		77,337 (5.2)		25	30	36	
	≤29	8940 (0.6)	233	290	306	316	
	29–≤ 32	12,995 (0.9)	106	180	189	195	
	<32	15,898 (1.1)	195	281	290	294	
	32–≤36	95,439 (6.4)	59	96	103	120	
	36	46,955 (3.2)	40	72	96	108	125
	>36	1,370,695 (92.1)	19	36	46	52	63
	37	109,343 (7.3)	28	46	61	64	81
	>37	1,261,352 (84.8)	18	35	45	51	59
≤2500	<32	15,319 (1.0)	205	283	292	307	
≤2500	32–≤36	55,237 (3.7)		105	123	147	
≤2500	>36	49,535 (3.3)		62	70	85	
≤2500	36	18,563 (1.2)	40	72	89	101	
>2500	36	28,392 (2.0)	38	70	90	109	
>2500	37	88,219 (5.9)		48	59	65	
>2500	≤37	129,194 (8.7)		47	62	63	71
>2500	>37	1,232,941 (82.9)		25	29	34	43

These data include both the reported cohort and an earlier pilot cohort of 109,937 newborns.

[a]Shown are 17a-hydroxyprogesterone values (nmol/L in blood) for different percentiles stratified by birth weight (BW) alone, gestational age (GA) alone, and both birth weight and gestational age.

Table 2. Israeli Newborn Screening Program Algorithm for 17-OHP

BW (g)	GA (wk)[a]	Repeat Requested Cutoff[b]	Cutoff for Referral Using First Sample[b]	Recall Sample Is Normal If:[c]
≤2500	<32	180		Reduction > 30%
≤2500	≥32 but ≤36	105		Reduction > 30%
≤2500	>36 or unknown	85		Reduction > 30%
>2500	≤37	70	90	< 70
>2500	>37 or unknown	35[d]	90	< 35

The cutoff level of 17-OHP is adjusted according to BW and GA.

[a]GA is rounded up to the higher wk (*i.e.*, 36 + 1 is considered 37 wks).

[b]Values in nmol/L in blood.

[c]Reduction compared with previous sample.

[d]Reduced in 2013 from 40 (99.99th centile) to 35 (99.98th centile) nmol/L in blood.

salt-wasting type CAH, who was referred to the emergency department following NBS results at 14 days of life). During the first 2 years, as the database was compiled, adjustments were made for the cutoff values in each percentile.

With regard to BW, we observed that a BW of 2500 g differentiates between a small minority of newborns (8.2%) and the majority. No significant differences were noted between very low BW (<1500 g) to early prematurity (≤29 weeks of gestation). From gestational age alone, we observed that for gestational ages < 32 weeks and < 29 weeks, the cutoff values would be similar at 280 nmol/L, whereas for gestational ages 29 to 32 weeks, a significantly different cutoff would be suitable, at 180 nmol/L. Thus, we decided to refer to a single age group of <32 weeks, maintaining the cutoff of 180 nmol/L, which is consistent with the 99th percentile. The consideration behind this decision was the small number of newborns within this category (approximately 15 cases per year), who would in any case be hospitalized in NICUs and thus for whom a repeated sample would be easily available.

In addition, we observed a difference in cutoffs between a GA of 36 weeks and a GA older than 36 weeks. Combining both BW and GA revealed two additional subgroups: newborns born at > 36 weeks who had a BW of ≤2500 g (constituting 3.3% of newborns), and those born at ≤37 weeks but with a BW of >2500 g (constituting 8.7% of newborns). A distinct cutoff level adjusted for the different groups decreased many of the cases that would have otherwise been designated as FP results. The number of infants in each GA, BW, and combined BW and GA categories are detailed in Table 1 and Fig. 1.

With regard to percentiles, for the smaller subgroups (*i.e.*, preterm newborns), relatively lower percentiles were used, such as the 99th centile or 99.9th centile. In contrast, for the larger subgroups we used a cutoff consistent with the 99.97th to 99.99th centiles.

Statistical analysis

Sensitivity was calculated as the proportion of cases with true-positive findings divided by the sum of those with true-positive and false-negative (FN) findings. Specificity was calculated as the proportion of cases with true-negative findings divided by the sum of those with true negative and false-positive findings. Positive predicted value was calculated as the proportion of cases with true-positive findings divided by the sum of those with true-positive and false-positive findings. We used the χ^2 test to compare the difference in recall rates between preterm and full-term infants. The 17-OHP values in Table 1 were expressed as percentiles. Statistical significance was set at $P < 0.05$.

Results

Between January 1, 2008, and December 31, 2017, a total of 1,378,132 newborns were documented and included in the national Israeli Newborn Screening Program (Fig. 2). Of these, 1,248,908 (91%) were admitted to the nursery, and 129,224 (9%) were admitted to the NICU; 8.5% of newborns were born preterm. A total of 1,033,599 (75%) of newborns were of Jewish descent, 303,189 (22%) were of Arab-Muslim descent, 18181 (1.3%) were of Arab-Christian descent, 4327 (0.3%) were non-Jewish European whites, and 7484 (0.5%) were of non-Jewish African descent.

True-positives

A total of 88 newborns were diagnosed with CAH, 50 (57%) of whom were male and 38 (43%) of whom were female. Fifty-three (60%) of the patients with proved CAH were of Jewish descent, yielding an incidence of 1 per 20,500 live births in the Jewish population (excluding 3 who were a second affected siblings in their family); 34 (39%) were of Arab-Muslim descent (2 of whom were the second affected child in their families), yielding an incidence of 1 per 9500 live births in the Arab population; and 1 (1%) was of European-Christian descent (Table 3).

Eighty-four of the 88 true-positive newborns with CAH were detected by using the aforementioned NBS protocol. Clinically, of the 88 patients with CAH, 59 (67%) were diagnosed with the salt-wasting form of CAH, 9 (10%) had the simple virilizing form, 1 had 11β-hydroxylase deficiency, 2 (siblings) had 3β-hydroxysteroid dehydrogenase deficiency, 5 (6%) had the nonclassic form, and the status of the additional 12 (14%) was unknown in this regard.

Among the 50 males diagnosed, in 43 (86%) the initial suspicion of CAH was raised following the NBS results. Of these, 6 had a positive family history of CAH that not

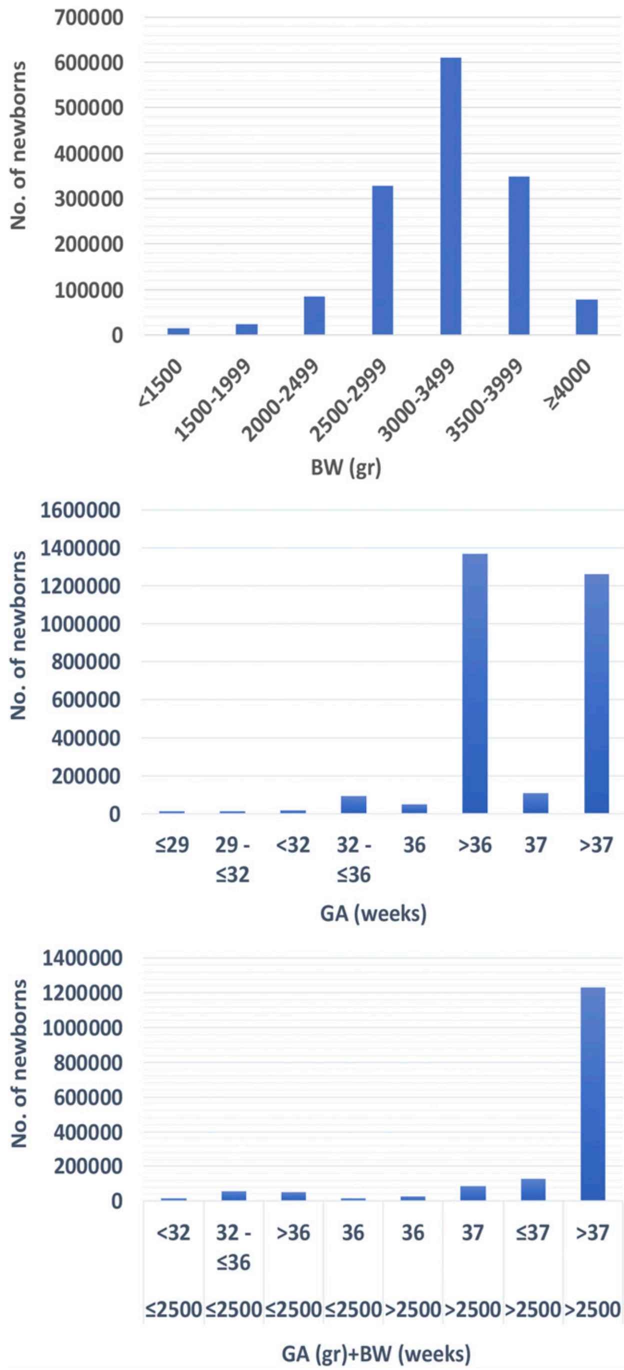

Figure 1. Histograms of the number of infants in each GA, BW, and combined BW and GA categories.

been reported upon medical history-taking before birth. The other 7 (14%) were diagnosed prior to NBS results: For 5 a family history of CAH had been reported, 1 had cryptorchidism and hypospadias (no uterus), and 1 was born outside of Israel and was diagnosed at age 2 months.

Of the 38 females diagnosed, 25 were recognized at birth with ambiguous genitalia, 4 were initially designated as male and later recognized as female, and 4 were diagnosed based on family history of CAH. The additional 5 affected females were diagnosed only following the NBS results: One of these had 3 β-hydroxysteroid dehydrogenase deficiency (25), in 2 ambiguous genitalia was not recognized at birth, and 2 had nonclassic CAH.

False-positive Cases

During the study period, 425 positive cases (0.03%) were designated as FP results (recall sample found within normal NBS limits). These included 227 newborns for whom DBS samples were obtained in NICUs (0.18% of all NICU newborns), and 198 newborns for whom the first DBS samples were obtained in the nursery (0.016%).

False-negative cases

Of the 88 newborns ultimately diagnosed with CAH, 4 were missed by the NBS program and were thus FN cases (Table 4). Three of these newborns were female and presented with ambiguous genitalia. One male was diagnosed shortly after birth following a positive family history. All four were ultimately diagnosed with the simple virilizing type of CAH.

Following our experience with these four FN cases, we aimed to further investigate whether other male newborns had died of CAH and had not been detected by the NBS program. For this purpose, we searched the National Registrar of Deceased for the study period (2008 through 2017) and found 1690 neonates who died between 5 and 61 days of age. For all these, we retrieved data on GA, BW, sex, and the 17-OHP level at first DBS sample. These cases were further reduced to include males with a GA of 35 weeks or older, BW of >2499 g, and a 17-OHP level of ≥30 nmol/L. Based on these criteria, 6 cases were left. Their clinical characteristics are summarized in Table 5. Importantly, none of these cases were ultimately diagnosed with classic salt-wasting CAH.

Discussion

This large prospective study used combined GA- and BW-adjusted cutoffs on 1,378,132 newborns screened for CAH. The PPV (16.5%) was high despite the low recall rate (0.03%). Sensitivity and specificity of the NBS for CAH were 95.4% and 99.9%, respectively, and the percentage of true-positive cases from all newborns referred for evaluation following a positive NBS result was 96%.

Collective global experience with NBS for CAH according to cutoff values based on BW or GA shows relatively high FP rates, hampering the success of NBS for this group of disorders. In 1998, a 1-year pilot study from the Netherlands was reported by van der Kamp *et al.* (26), in which the 17-OHP cutoff levels further stratified by GA were met only for newborns with BW ≤ 2500 g.

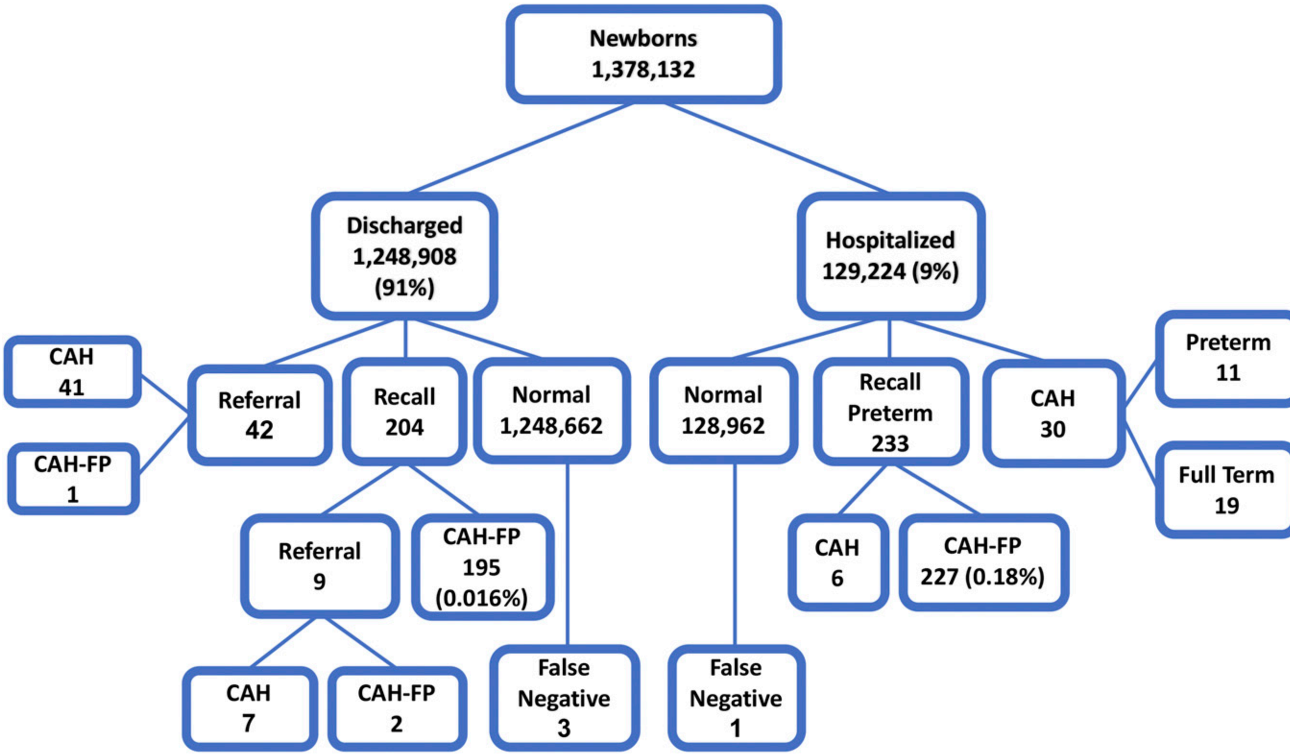

Figure 2. Algorithm describing results of the NBS for CAH in Israel from January 2008 to December 2017.

The prevalence in 1998 was 1 per 8783. However, in 1999, the second year of the pilot study, the cutoff was based on GA alone, with a prevalence of 1 per 17,771, which raises the concern of unrecognized FN cases. In a retrospective population-based study in Tokyo, Japan, the cutoff level was adjusted for GA and BW (8). The PPV was 25.8%, but no data were reported on other screening parameters. Characteristics of the respective experiences in five NBS programs is summarized in Table 6.

Linder *et al.* (27), using serial measurements of 17-OHP in an Israeli cohort of 66 consecutive preterm newborns (and 100 term neonates, serving as controls), demonstrated that, indeed, these levels are significantly higher in preterm infants and highest in postconceptional age <29 weeks. Moreover, they showed that for infants born at GA >31 weeks, 17-OHP levels dropped over time and reached those seen in term infants by 3 months of age. This prompted us to pursue a strategy where in addition to cutoff levels based on both GA and BW, a decrease of at least 30% in the 17-OHP level in a recall sample was defined as a negative screening result, and thus deemed a false positive case. Similar determination was also reported by Gidlöf *et al.* (7), where a reduction of over 20% and no clinical signs for a recall sample was classified as false positive results.

Our 10 years of experience using cutoffs based on the combination of these two factors show that this strategy enables higher cutoff values, subsequently reducing the FP rates to as low as 0.03% and thus making this strategy superior to those previously described. The International Society of Newborn Screening recommends, based on GA (week) alone, the use of 99.9th percentile for ≥36 weeks and the 99.5th percentile for preterm infants (28). Gidlöf and colleagues' study demonstrated 0.06% (0.57%

Table 3. Demographic Data and CAH Detection Rates Among Newborns Included in the Israeli Newborn Screening Program Between January 1, 2008, and December 31, 2017

Ethnic Origin	Percentage of Newborns	Newborns With CAH	Incidence[a]
Jewish	75% (n = 1,033,599)	53 (3 were second affected child in family)	1 per 20,500
		4 of the 53 were of Ethiopian-Jewish origins[b]	1 per 8000
Arab-Muslim	22% (n = 303,189)	35 (2 were second affected child in family)	1 per 9000

[a]Excluding the second affected sibling.

[b]In Ethiopian Jews, the incidence of CAH was 1:8000 (based on data obtained from the Israeli Ministry of Interior, according to which 2.3% of all Israeli newborns in the last decade were born to parents born in Ethiopia).

Table 4. FN Cases Reported in Newborns Born Between January 1, 2008, and December 31, 2017

FN Case	Sex	BW[a]	GA[a]	17-OHP (nmol/L in Blood)	Clinical Type of CAH	Additional Information
1	Female	4396	40	34.0	SV	Ambiguous genitalia; would have been missed by both cutoffs: 40 and 35
2	Female	2620	36	56 169	SV	Ambiguous genitalia; second sample from NICU
3	Female	2290	39	38.2	SV	Ambiguous genitalia; cutoff was 40 nmol/L in blood
4	Male	4040	41	18.9 17.3	SV	Maternal treatment with glucocorticoids from GA 6 to 13 Sample was retested after diagnosis

Abbreviation: SV, simple virilizing.

[a]BW in grams; GA is rounded up to the higher wk (i.e. 36 + 1 is considered 37 wk).

preterm infants and 0.03% in full-term infants) FP. which is the closest to our low FP rate, however, they reported a very low sensitivity with 42 FN cases (sensitivity of 84.3%) (7). Our results also showed FP rates 10-fold higher in the preterm vs the term group (0.18% vs 0.016%, respectively) with overall 0.03% FP rates, but only 4 FN cases (sensitivity of 95.4%). These few FN cases raised the question of whether additional male cases might have gone unrecognized. Subsequently, we revisited the clinical diagnoses of all deceased males who could have potentially been such FN results (Table 5). Because none of these were ultimately diagnosed with classic salt-wasting CAH, these results demonstrate the safety of using the 99.97th percentile for full-term newborns.

In our large cohort there were, to our knowledge, four FN cases. Of these, three were females diagnosed because of ambiguous genitalia, and one male was diagnosed on the basis of a family history (Table 4). One of the FN results was a case of a female newborn (designated no. 3 in Table 4) initially suspected to have the salt-wasting type, found to be healthy upon recall to the hospital, only to be later genetically diagnosed with the simple virilizing type of CAH. After this case, and through continuous monitoring of reported cases, the cutoff level was lowered in 2013. Another of the FN cases (designated no. 4 in Table 4) showed levels well within the normal range, attributable to maternal glucocorticoid treatment during pregnancy, a known cause of FN results. Interestingly, all three relevant studies compared in Table 6 report that all FN cases were of the simple virilizing type of CAH.

The most striking differences seen in our study concerned the number of newborns referred for re-evaluation and, more importantly, the percentage of true-positive cases from within this group (4% and 26% vs 96%) (Table 6). Although one must consider the earlier sampling performed in New York State (29), these findings demonstrate a substantial advantage to the newly described strategy with regard to both cost-effectiveness and the reduction of unnecessary recalls and their associated emotional strain on the parents.

The Israeli population has unique demographic characteristics, which are relevant to the interpretation of the NBS data. First, it is extremely diverse with regard to ethnic background, with numerous ethnic groups, each with its carrier rates of inherited disorders. Second, several ethnic groups (for instance, the Orthodox-Jewish population and parts of the Arab-Muslim populations) are characterized by marriages within their own communities and, in some, by high rates of consanguinity, and hence have a higher risk of autosomal recessive diseases (30). Indeed, our data showed

Table 5. Clinical Characteristics of Six Deceased Infant Males With Blood 17-OHP Level > 30 nmol/L

Case	GA[a]	BW[a]	17-OHP (nmol/L in Blood)	Age at Death (d)	NBS Comment	Diagnoses
1	38	2565	49.5	5	Recall for CAH was requested	Hypoxic ischemic encephalopathy of newborn
2	36	2630	32.0	6	Recall for amino acids was requested	Ebstein anomaly/opening of tricuspid valve is displaced toward the apex of the right ventricle
3	39	3330	30.9	6		Sepsis of newborn due to anaerobes
4	41	3894	40.7	9	Retest showed 31.3 nmol/L blood	Ornithine transcarbamylase deficiency
5	35	2570	35.7	9		Potter syndrome
6	39	2770	30.1	20		Severe birth asphyxia

[a]BW in grams; GA is rounded up to the higher week (*i.e.*, 36 + 1 is considered 37 wks).

Table 6. Characteristics of Experiences of Newborn Screening Programs for CAH in the Netherlands, Japan, New York State, Sweden, and Israel

Characteristic	Netherlands, 1998	Netherlands, 1999	Japan	New York	Sweden	Israel
Newborns, n	87,827	88,857	2,105,108	1,962,433	2,737,932	1,378,132
Repeat requested	214	26	NA	10,574	1,728	425
FP, %	0.24	0.03	NA	0.54	0.06	0.03
Referred, n	NA	NA	410	2476	NA	87
True-positive, n	10	5	106	105	231	84
True-positive from referred, %	NA	NA	26	4	NA	96
FN cases, n	0	0	NA	3	43	4
Sensitivity, %	100	100	NA	98.5	84.3	95.4
Specificity, %	99.96	99.97	NA	99.9	99.9	99.9
Overall PPV, %	4.5	16	NA	5.8	13.4	16.5
PPV for full-term newborns, %	NA	NA	NA	12.6	25.1	19.5
Cutoff values GA/BW/GA + BW[a]	BW	GA	GA/BW	BW	GA	GA+BW

For location-specific characteristics, see references 7, 8, 26, and 29.

[a]Cutoff values for 17-OHP were determined according to GA or BW or a combination of both GA and BW (GA + BW).

that the incidence of CAH in the Arab-Muslim population is 1 per 9000, similar to that reported in other ethnic groups with relatively high consanguinity rates (31). Interestingly, an unexpected finding of the NBS experience was the relatively high incidence of CAH within the Ethiopian-Jewish population, reaching as high as 1 per 8000 (Table 3). This was surprising because it had not been previously described in the medical literature and the Ethiopian-Jewish community typically avoids consanguineous marriages (32). The high incidence within this population was not observed by the endocrinologists scattered throughout Israel but only when the data were summed at the national level. Hence, the current study identifies an at-risk population not previously recognized.

To conclude, we suggest that the use of cutoff values of 17-OHP levels based on a combination of GA and BW for newborn screening of classic CAH is safe and more efficient than either factor alone. This strategy enables lower FP rates and reduces the costs and emotional strain accompanying unnecessary recalls and referrals of FP cases.

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